1. Aggregate Functions

Aggregate functions **summarize numeric data** (total, average, maximum, etc.).
Example Table: **Sales**

Product	Amount
A	100
B	200
C	300

SUM

Adds all values in a column.
Total Sales = SUM(Sales[Amount])
Result
100 + 200 + 300 = **600**
Use Case
Total revenue, total profit, total quantity.

MAX

Returns the **largest value**.
Max Sale = MAX(Sales[Amount])
Result = **300**

MIN

Returns the **smallest value**.
Min Sale = MIN(Sales[Amount])
Result = **100**

AVERAGE

Calculates the **average value**.
Average Sales = AVERAGE(Sales[Amount])
Result
(100 + 200 + 300) / 3 = **200**

COUNT

Counts **numeric values** in a column.
Total Transactions = COUNT(Sales[Amount])
Result = **3**

---

2. Logical Functions

Logical functions are used for **conditions and decision making**.

IF

Checks a condition and returns one value if true and another if false.
Example

Amount
600
200

Formula
Category =
IF(Sales[Amount] > 500 , "High" , "Low")
Result

Amount	Category
600	High
200	Low

AND

Checks **multiple conditions**.

IF(
AND(Sales[Amount] > 200 , Sales[Amount] < 500),
"Medium",
"Other"
)

Meaning

Value must satisfy **both conditions**.

OR

Returns TRUE if **any condition is true**.

IF(
OR(Sales[Region]="East", Sales[Region]="West"),
"Priority",
"Normal"
)

NOT

Reverses the logical condition.

IF(
NOT(Sales[Amount] > 500),
"Below Target",
"Above Target"
)

SWITCH

Used instead of multiple IF statements.

Grade =
SWITCH(
TRUE(),
Sales[Amount] > 500 , "A",
Sales[Amount] > 300 , "B",
Sales[Amount] > 100 , "C",
"D"
)

---

3. Filter Functions

Filter functions **modify the filter context** in Power BI.

CALCULATE

The **most important DAX function**.
It **changes filter context**.
Example Table

Region	Amount
East	100
West	200
East	300

Formula

East Sales =
CALCULATE(
SUM(Sales[Amount]),
Sales[Region] = "East"
)

Result

100 + 300 = **400**

FILTER

Returns a **filtered table based on condition.**

High Sales =
FILTER(
Sales,
Sales[Amount] > 200
)

Result table

Product	Amount
C	300

ALL

Removes all filters from a table.

Total Sales All =
CALCULATE(
SUM(Sales[Amount]),
ALL(Sales)
)

Meaning

Ignore all slicers and filters.

ALLEXCEPT

Removes filters **except specific columns**.

CALCULATE(
SUM(Sales[Amount]),
ALLEXCEPT(Sales, Sales[Region])
)

Meaning

Keep **Region filter** but remove others.

ALLSELECTED

Removes filters from visual but **keeps slicer filters**.

CALCULATE(
SUM(Sales[Amount]),
ALLSELECTED(Sales)
)

---

4. Iterator Functions
Iterator functions calculate **row by row**.
They usually end with **X**.

SUMX
Example Table

Product	Qty	Price
A	2	100
B	3	200

Formula
Total Sales =
SUMX(
Sales,
Sales[Qty] * Sales[Price]
)
Calculation
2 × 100 = 200
3 × 200 = 600
Total = **800**

AVERAGEX
AVERAGEX(
Sales,
Sales[Qty] * Sales[Price]
)
Average of calculated values.

MAXX
MAXX(
Sales,
Sales[Qty] * Sales[Price]
)
Returns the **maximum calculated value**.

---

5. Text Functions
Used to manipulate **text strings**.

LEFT
Extract characters from the **left side**.
LEFT(Customer[Name],3)
Rahul → **Rah**

UPPER
Converts text to **uppercase**.
UPPER(Customer[Name])
rahul → **RAHUL**

LOWER
Converts text to **lowercase**.
LOWER(Customer[Name])
RAHUL → **rahul**

MID
Extract characters from **middle**.
MID(Customer[Name],2,3)
Rahul → **ahu**

CONCATENATE
Full Name =
CONCATENATE(Customer[FirstName],Customer[LastName])
Rahul + Sharma → **RahulSharma**

LEN
Counts number of characters.
LEN(Customer[Name])
Rahul → **5**

---

6. Date & Time Functions

YEAR
YEAR(Sales[Date])
Example
2024-06-10 → **2024**

MONTH
MONTH(Sales[Date])
Result = **6**

DAY
DAY(Sales[Date])
Result = **10**

TODAY
TODAY()
Returns **current date**.

NOW
NOW()
Returns **current date and time**.

---

7. Time Intelligence Functions
Used for **year-to-date, month-to-date analysis**.
Important: requires a **Date table**.

TOTALYTD
Sales YTD =
TOTALYTD(
SUM(Sales[Amount]),
Date[Date]
)
Returns **sales from start of year until today**.

TOTALMTD
TOTALMTD(
SUM(Sales[Amount]),
Date[Date]
)
Returns **month-to-date sales**.

TOTALQTD
TOTALQTD(
SUM(Sales[Amount]),
Date[Date]
)
Returns **quarter-to-date sales**.

SAMEPERIODLASTYEAR
Sales Last Year =
CALCULATE(
SUM(Sales[Amount]),
SAMEPERIODLASTYEAR(Date[Date])
)
Compares **current period vs last year**.

NEXTMONTH / PREVIOUSMONTH
NEXTMONTH(Date[Date])
Returns next month dates.
PREVIOUSMONTH(Date[Date])
Returns previous month dates.

---

8. Relationship Functions

RELATED
Gets value from **related table**.
Customer Table

CustomerID	Name
1	Rahul

Sales Table

CustomerID	Amount
1	500

Formula
Customer Name =
RELATED(Customer[Name])
Result = **Rahul**

RELATEDTABLE
Returns related rows from another table.
Order Count =
COUNTROWS(RELATEDTABLE(Sales))

USERELATIONSHIP
Activates an **inactive relationship**.

CALCULATE(
SUM(Sales[Amount]),
USERELATIONSHIP(Sales[ShipDate],Date[Date])
)

---

9. Most Important DAX Functions for Data Analysts
These are the **top 10 functions used in most Power BI dashboards**.

SUM
SUM(Sales[Amount])

CALCULATE
CALCULATE(
SUM(Sales[Amount]),
Sales[Region] = "East"
)

FILTER
FILTER(Sales, Sales[Amount] > 500)

SUMX
SUMX(Sales, Sales[Qty] * Sales[Price])

ALL
ALL(Sales)

DISTINCTCOUNT
DISTINCTCOUNT(Sales[CustomerID])
Counts **unique customers**.

RANKX
Rank Product =
RANKX(
ALL(Sales[Product]),
SUM(Sales[Amount])
)
Ranks products by sales.

DIVIDE
Profit Margin =
DIVIDE(
SUM(Sales[Profit]),
SUM(Sales[Amount])
)
Safe division (prevents divide by zero).
